

# DIFFERENTIALS, DUAL ISOGENIES, AND THE TATE MODULE: SOLUTIONS

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Three computations by hand, on the invariant differential, the dual isogeny and the Tate module, each comparing the multiplicative group  $\mathbb{G}_m$  with the elliptic curve

$$E: y^2 = x^3 - x, \quad \omega_E := \frac{dx}{2y}.$$

Throughout,  $k$  is a field with  $\text{char } k \neq 2$ ,  $\bar{k}$  an algebraic closure and  $k^{\text{sep}} \subseteq \bar{k}$  the separable closure;  $\mathbb{G}_m = \mathbb{P}_k^1 \setminus \{0, \infty\}$  has coordinate  $t$  and invariant differential  $\omega_{\mathbb{G}_m} := dt/t$ .

**Reading.** [Sil09, §§III.4–III.7], and [Sil09, §III.8] for the Weil pairing of Fact 3.1.

**A formula.** We use the following duplication formula on  $E$ , derived in Problem 1.1(5): if  $x \in k(E)$  is the  $x$ -coordinate and  $[2]: E \rightarrow E$  is multiplication by 2, then

$$(0.1) \quad [2]^*x := x \circ [2] = \frac{(x^2 + 1)^2}{4(x^3 - x)}.$$

## 1. INVARIANT DIFFERENTIALS

**Problem 1.1** (Poles and pullbacks).

- (1) Rewrite  $\omega_{\mathbb{G}_m}$  in the coordinate  $s := 1/t$  near  $\infty$ , and compute  $\text{div}(\omega_{\mathbb{G}_m})$  on  $\mathbb{P}_k^1$ . Compute  $[m]^*\omega_{\mathbb{G}_m}$  for  $[m](t) = t^m$ ,  $m \in \mathbb{Z}$ .
- (2) Show that  $\omega_E = dy/(3x^2 - 1)$ , and that  $y$  and  $3x^2 - 1$  have no common zero on  $E$ . Deduce that  $\omega_E$  has no zeros or poles on  $E \setminus \{O\}$ .
- (3) Using  $\text{ord}_O x = -2$  and  $\text{ord}_O y = -3$ , show that  $\text{ord}_O \omega_E = 0$ .
- (4) Compute  $[-1]^*\omega_E$  from  $[-1](x, y) = (x, -y)$ , and compare with  $[-1]^*\omega_{\mathbb{G}_m}$ .
- (5) We derive (0.1) by hand. Here  $x, y \in k(E)$  are functions; for a point  $P \in E(\bar{k})$  write  $x_P := x(P)$  and  $y_P := y(P)$  for its coordinates.
  - (a) Let  $P \in E(\bar{k})$  with  $y_P \neq 0$ . Show that the tangent line to  $E$  at  $P$  has slope  $\lambda_P = (3x_P^2 - 1)/(2y_P)$ , and use that its third point of intersection with  $E$  is  $-[2]P$  to show that  $x([2]P) = \lambda_P^2 - 2x_P$ .
  - (b) Let  $\lambda := (3x^2 - 1)/(2y) \in k(E)$ , so that  $\lambda(P) = \lambda_P$ . Explain why (5)(a) gives the identity  $[2]^*x = \lambda^2 - 2x$  of functions in  $k(E)$ , and simplify it using  $y^2 = x^3 - x$ .
- (6) Solve this part in **Magma**, working in the function field  $\mathbb{Q}(E) = \mathbb{Q}(x)[y]/(y^2 - x^3 + x)$ . Compute  $[2]^*x$  and  $[2]^*y$  from the function  $\lambda$  of (5), check (0.1), and show that

$$[2]^*x - e = \frac{g_e(x)^2}{4(x^3 - x)}, \quad g_0 = x^2 + 1, \quad g_1 = x^2 - 2x - 1, \quad g_{-1} = x^2 + 2x - 1,$$

for  $e \in \{0, 1, -1\}$ . Show that  $[2]^*y = \pm(x^2 + 1)(x^4 - 6x^2 + 1)/(8y^3)$  and  $[2]^*\omega_E = \pm 2\omega_E$ . Which sign is the correct one?

*Solution.*

(1)  $t = 1/s$  gives  $dt = -ds/s^2$ , so  $\omega_{\mathbb{G}_m} = -ds/s$ . Hence  $\omega_{\mathbb{G}_m}$  has simple poles at 0 and  $\infty$ , with residues 1 and  $-1$ , and no other zeros or poles:  $\text{div}(\omega_{\mathbb{G}_m}) = -(0) - (\infty)$ . Also  $[m]^*\omega_{\mathbb{G}_m} = d(t^m)/t^m = m\omega_{\mathbb{G}_m}$ .

(2) Differentiating  $y^2 = x^3 - x$  gives  $2y dy = (3x^2 - 1) dx$ . If  $y = 0$ , then  $x \in \{0, \pm 1\}$  and  $3x^2 - 1 \in \{-1, 2\}$ , which is non-zero. On the open set  $y \neq 0$  the differential  $dx$  generates  $\Omega_{E/k}$ , and  $\omega_E = dx/(2y)$  is a unit times it; on  $3x^2 - 1 \neq 0$ , likewise with  $dy$  and  $dy/(3x^2 - 1)$ . These open sets cover  $E \setminus \{O\}$ .

(3) With a uniformizer  $w$  at  $O$ , write  $x = w^{-2}u$  with  $u(O) \neq 0$ . Then  $dx = (-2w^{-3}u + w^{-2}u')dw$  has order  $-3$ , because  $2 \neq 0$  in  $k$ . So  $\text{ord}_O \omega_E = -3 - (-3) = 0$ .

(4)  $[-1]^*\omega_E = dx/(-2y) = -\omega_E$ . On  $\mathbb{G}_m$ ,  $[-1](t) = 1/t$  and  $[-1]^*\omega_{\mathbb{G}_m} = -\omega_{\mathbb{G}_m}$  as well.

(5)(a) In the affine plane with coordinates  $x, y$ , differentiating  $y^2 = x^3 - x$  at  $P$  gives  $2y_P dy = (3x_P^2 - 1) dx$ , so the tangent line at  $P$  is  $y = \lambda_P x + \nu_P$ , with  $\lambda_P = (3x_P^2 - 1)/(2y_P)$  and  $\nu_P = y_P - \lambda_P x_P$ . Substituting into  $y^2 = x^3 - x$  gives

$$x^3 - \lambda_P^2 x^2 - (1 + 2\lambda_P \nu_P)x - \nu_P^2 = 0,$$

whose roots are  $x_P$ , twice because the line is tangent, and  $x(-[2]P)$ . Comparing coefficients of  $x^2$  gives  $x(-[2]P) = \lambda_P^2 - 2x_P$ , and  $x(-[2]P) = x([2]P)$  since  $[-1](x, y) = (x, -y)$ .

(5)(b) By definition of the pullback,  $([2]^*x)(P) = x([2]P)$ , and  $(\lambda^2 - 2x)(P) = \lambda_P^2 - 2x_P$ . So by (5)(a) the function  $[2]^*x - (\lambda^2 - 2x) \in k(E)$  vanishes at every  $P \in E(\bar{k})$  with  $y_P \neq 0$ , that is, at every point of  $E(\bar{k})$  outside the finite set  $E[2](\bar{k})$ . A non-zero rational function has only finitely many zeros, so  $[2]^*x = \lambda^2 - 2x$  in  $k(E)$ . This is the precise meaning of the shorthand “ $x([2]P) = \lambda^2 - 2x$  for  $P = (x, y)$ ”. Finally, in  $k(E)$  we have  $4y^2 = 4(x^3 - x)$ , so

$$[2]^*x = \lambda^2 - 2x = \frac{(3x^2 - 1)^2 - 8x(x^3 - x)}{4(x^3 - x)} = \frac{x^4 + 2x^2 + 1}{4(x^3 - x)} = \frac{(x^2 + 1)^2}{4(x^3 - x)}.$$

(6) The identities have coefficients in  $\mathbb{Z}[1/2]$ , so it suffices to check them over  $\mathbb{Q}$ . By (5),  $[2]^*x = \lambda^2 - 2x$  with  $\lambda = (3x^2 - 1)/(2y)$ , and the same argument applied to the  $y$ -coordinate of the third point of intersection gives  $[2]^*y = -(y + \lambda([2]^*x - x))$ . Since  $\omega_E = dx/(2y)$ , we have  $[2]^*\omega_E/\omega_E = (d([2]^*x)/dx)/(2[2]^*y) \cdot 2y$ .

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Kx<x> := FunctionField(Rationals());
R<t> := PolynomialRing(Kx);
F<y> := ext<Kx | t^2 - (x^3 - x)>; // q(E)
lam := (3*x^2 - 1)/(2*y); // tangent slope
x2 := lam^2 - 2*x; y2 := -(y + lam*(x2 - x)); // [2]^*x, [2]^*y
x2 eq (x^2 + 1)^2/(4*(x^3 - x)); // true
[ Factorization(Numerator(Kx!(x2 - e))) : e in [0, 1, -1] ];
// [ [ <x^2 + 1, 2> ], [ <x^2 - 2*x - 1, 2> ], [ <x^2 + 2*x - 1, 2> ] ]
y2 eq (x^2 + 1)*(x^4 - 6*x^2 + 1)/(8*y^3); // true
(F!Derivative(Kx!x2))/(2*y2) * 2*y; // 2
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The denominator of  $[2]^*x - e$  is that of  $[2]^*x$ , namely  $4(x^3 - x)$ , and the factorizations show that its numerator is  $g_e(x)^2$ . Both signs are  $+$ :  $[2]^*y = (x^2 + 1)(x^4 - 6x^2 + 1)/(8y^3)$  and  $[2]^*\omega_E = 2\omega_E$ , as [Sil09, Corollary III.5.3] predicts.

Alternatively, Magma’s elliptic curve functionality provides  $[2]$  as an isogeny, and pulls back both functions and differentials along it, so no group law is typed in.

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E := EllipticCurve([-1,0]); // y^2 = x^3 - x
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m2 := MultiplicationByMMap(E, 2);           // [2] : E -> E
KE<x,y> := FunctionField(E);               // Q(E)
x2 := Pullback(m2, x); y2 := Pullback(m2, y); // [2]^*x, [2]^*y
x2 eq (x^2 + 1)^2/(4*(x^3 - x));           // true
g := [x^2 + 1, x^2 - 2*x - 1, x^2 + 2*x - 1]; // g_0, g_1, g_{-1}
[ x2 - [0, 1, -1][i] eq g[i]^2/(4*(x^3 - x)) : i in [1..3] ];
// [ true, true, true ]
y2 eq (x^2 + 1)*(x^4 - 6*x^2 + 1)/(8*y^3); // true
omega := Differential(x)/(2*y);           // omega_E
Pullback(m2, omega)/omega;               // 2

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## 2. DUAL ISOGENIES

Consider the elliptic curves

$$E_1: y_1^2 = x_1^3 - x_1, \quad E_2: y_2^2 = x_2^3 + 4x_2, \quad E_3: y_3^2 = x_3^3 - 16x_3,$$

so that  $E_1 = E$ , and write  $\omega_i := dx_i/(2y_i)$  for the invariant differential of  $E_i$ . Consider the maps

$$\begin{aligned} \varphi: E_1 &\rightarrow E_2, \quad (x_1, y_1) \mapsto \left( \frac{x_1^2 - 1}{x_1}, \frac{y_1(x_1^2 + 1)}{x_1^2} \right), \\ \psi: E_2 &\rightarrow E_3, \quad (x_2, y_2) \mapsto \left( \frac{y_2^2}{x_2^2}, \frac{y_2(x_2^2 - 4)}{x_2^2} \right), \\ \iota: E_3 &\rightarrow E_1, \quad (x_3, y_3) \mapsto \left( \frac{x_3}{4}, \frac{y_3}{8} \right). \end{aligned}$$

For an isogeny  $\alpha: E_i \rightarrow E_j$ , define  $c_\alpha$  by  $\alpha^*\omega_j = c_\alpha\omega_i$ . The maps  $\varphi$  and  $\psi$  are special cases of [Sil09, Example III.4.5], which treats  $y^2 = x^3 + ax^2 + bx$  in general, so they are isogenies;  $\iota$  is an isomorphism. That example computes the kernels and the dual, but not the constants  $c_\alpha$ : that is part (2) of Problem 2.1.

**Problem 2.1** (A 2-isogeny and its dual).

- (1) Check that  $\varphi$  lands on  $E_2$ . Show that  $\ker \varphi = \{O_1, (0, 0)\}$ , that  $\varphi(\pm 1, 0) = (0, 0)$ , and that  $\deg \varphi = 2$ .<sup>a</sup>
- (2) Compute  $c_\varphi$ .
- (3) Check that  $\psi$  lands on  $E_3$  and that  $\iota$  lands on  $E_1$ , and compute  $c_\psi$  and  $c_\iota$ .
- (4) Show that  $(\iota \circ \psi \circ \varphi)^*x_1 = [2]^*x_1$ , so that  $\iota \circ \psi \circ \varphi = \pm[2]$ . Use (2) and (3) to decide the sign, and conclude that  $\hat{\varphi} = \iota \circ \psi$ , that  $c_\varphi c_{\hat{\varphi}} = \deg \varphi$ , and that  $\ker \hat{\varphi} = \varphi(E_1[2](\bar{k}))$ .
- (5) Is there an isogeny  $\mathbb{G}_m \rightarrow \mathbb{G}_m$  of degree 2? If so, does it have a dual?

<sup>a</sup>Hint: write  $y_1$  in terms of  $\varphi^*y_2$  and  $x_1$ , and find a quadratic equation for  $x_1$  over  $\varphi^*k(E_2)$ .

*Solution.*

(1) Since  $y_1^2 = x_1(x_1^2 - 1)$ , the functions  $\varphi^*x_2 = (x_1^2 - 1)/x_1$  and  $\varphi^*y_2 = y_1(x_1^2 + 1)/x_1^2$  satisfy

$$(\varphi^*y_2)^2 = \frac{x_1(x_1^2 - 1)(x_1^2 + 1)^2}{x_1^4}, \quad (\varphi^*x_2)^3 + 4\varphi^*x_2 = \frac{x_1^2 - 1}{x_1} \cdot \frac{(x_1^2 + 1)^2}{x_1^2},$$

which agree. The function  $\varphi^*x_2 = x_1 - 1/x_1$  has poles only at  $(0, 0)$  and at  $O_1$ , so  $\ker \varphi = \{O_1, (0, 0)\}$ ; and at  $(\pm 1, 0)$  both coordinates of  $\varphi$  vanish. Next,  $y_1 = x_1^2 \varphi^*y_2 / (x_1^2 + 1)$  and

$x_1^2 - (\varphi^*x_2)x_1 - 1 = 0$ , so  $k(E_1) = \varphi^*k(E_2)(x_1)$  has degree at most 2 over  $\varphi^*k(E_2)$ . Since  $\#\ker\varphi = \deg_s\varphi \leq \deg\varphi$ , we get  $\deg\varphi = 2$ .

(2)  $\varphi^*dx_2 = (1 + 1/x_1^2)dx_1 = (x_1^2 + 1)dx_1/x_1^2$ , so  $\varphi^*\omega_2 = \omega_1$  and  $c_\varphi = 1$ . This is why the second coordinate of  $\varphi$  has the sign it does: the opposite sign would give  $c_\varphi = -1$ .

(3)  $\varphi$  and  $\psi$  are the same construction for  $y^2 = x^3 + bx$ , with  $b = -1$  and  $b = 4$  respectively, and target  $y^2 = x^3 - 4bx$ ; the computation of (1) applies verbatim, and that of (2) gives  $c_\psi = 1$ . Also  $(y_3/8)^2 = (x_3^3 - 16x_3)/64 = (x_3/4)^3 - x_3/4$ , so  $\iota$  lands on  $E_1$ , and  $\iota^*\omega_1 = \frac{dx_3/4}{2y_3/8} = 2\omega_3$ , so  $c_\iota = 2$ .

(4) We have  $\psi^*x_3 = y_2^2/x_2^2 = x_2 + 4/x_2$  and  $\iota^*x_1 = x_3/4$ , so

$$(\iota \circ \psi \circ \varphi)^*x_1 = \frac{1}{4} \left( \frac{x_1^2 - 1}{x_1} + \frac{4x_1}{x_1^2 - 1} \right) = \frac{(x_1^2 + 1)^2}{4(x_1^3 - x_1)},$$

which is  $[2]^*x_1$  by (0.1). Two isogenies with the same pullback of  $x_1$  agree up to sign. Now  $c_{\iota \circ \psi \circ \varphi} = 2 \cdot 1 \cdot 1 = 2 = c_{[2]}$  by Problem 1.1(6), while  $c_{[-2]} = -2 \neq 2$ . So  $\iota \circ \psi \circ \varphi = [2]$ , and by uniqueness of the dual,  $\hat{\varphi} = \iota \circ \psi$ , with  $c_{\hat{\varphi}} = 2$  and  $c_\varphi c_{\hat{\varphi}} = 2 = \deg\varphi$ . Finally  $\ker\hat{\varphi} = \ker\psi = \{O_2, (0,0)\} = \varphi(E_1[2](\bar{k}))$  by (1).

(5) Yes. The endomorphisms of  $\mathbb{G}_m$  are the maps  $[m](t) = t^m$ ,  $m \in \mathbb{Z}$ , and  $\deg[m] = |m|$ , so the isogenies  $\mathbb{G}_m \rightarrow \mathbb{G}_m$  of degree 2 are  $[2]$  and  $[-2]$ , both with kernel  $\mu_2(\bar{k}) = \{\pm 1\}$ . What  $\mathbb{G}_m$  lacks is a dual: the only  $[n]$  with  $[n] \circ [m] = [\deg[m]]$  is  $[\text{sign } m]$ , and  $[m] \mapsto [\text{sign } m]$  is not additive. This fails because  $\deg[m] = |m|$ , not  $m^2$ .

### 3. THE TATE MODULE

We first describe the Tate module of  $\mathbb{G}_m$ , the familiar case, and then repeat each step for  $E$ . What links the two is the Weil pairing, which you may use without proof.

**Fact 3.1** (The Weil pairing). Let  $m \geq 1$  with  $\text{char } k \nmid m$ . There is a pairing

$$e_m : E[m](k^{\text{sep}}) \times E[m](k^{\text{sep}}) \rightarrow \mu_m(k^{\text{sep}})$$

that is bilinear, alternating ( $e_m(P, P) = 1$ ), non-degenerate, and Galois equivariant:  $\sigma(e_m(P, Q)) = e_m(\sigma P, \sigma Q)$  for every  $\sigma \in \text{Gal}_k$  [Sil09, Proposition III.8.1]. Here  $E[m](k^{\text{sep}}) = E[m](\bar{k})$ , since  $\text{char } k \nmid m$ . It follows that if  $(P, Q)$  is a basis of  $E[m](k^{\text{sep}})$  over  $\mathbb{Z}/m\mathbb{Z}$ , then  $e_m(P, Q)$  is a primitive  $m$ -th root of unity, and if  $\sigma$  acts on this basis by the matrix  $A \in \text{GL}_2(\mathbb{Z}/m\mathbb{Z})$ , then  $\sigma(e_m(P, Q)) = e_m(P, Q)^{\det A}$ .

**Problem 3.2** (The 2-power torsion of  $y^2 = x^3 - x$ ).

Regard  $E$  as a curve over  $\mathbb{Q}$ , fix  $\bar{\mathbb{Q}} \subset \mathbb{C}$ , and recall that  $E[m](\bar{\mathbb{Q}})$  is the kernel of  $[m]$  on  $E(\bar{\mathbb{Q}})$ . Let  $\sigma$  be complex conjugation, acting on  $\bar{\mathbb{Q}}$  and on  $E(\bar{\mathbb{Q}})$  coordinatewise.

- (1) Recall that  $T_2(\mathbb{G}_m) := \varprojlim_n \mu_{2^n}(\bar{\mathbb{Q}})$ , with transition maps  $\zeta \mapsto \zeta^2$ . Find a  $\mathbb{Z}_2$ -basis of  $T_2(\mathbb{G}_m)$ , and write down its first three terms. How does  $\sigma$  act on  $\mu_4(\bar{\mathbb{Q}})$ , and on  $T_2(\mathbb{G}_m)$ ?
- (2) List  $E[2](\bar{\mathbb{Q}})$ . Using Problem 1.1(6), find the  $x$ -coordinates of all  $P \in E(\bar{\mathbb{Q}})$  with  $[2]P \in E[2](\bar{\mathbb{Q}}) \setminus \{O\}$ , and conclude that  $\#E[4](\bar{\mathbb{Q}}) = 16$ .
- (3) Show that  $P_2 := (i, 1 - i)$  and  $Q_2 := (1 + \sqrt{2}, 2 + \sqrt{2})$  are points of  $E(\bar{\mathbb{Q}})$ , with  $[2]P_2 = (0, 0)$  and  $[2]Q_2 = (1, 0)$ . Show that  $(P_2, Q_2)$  is a basis of  $E[4](\bar{\mathbb{Q}})$  over  $\mathbb{Z}/4\mathbb{Z}$ , and that  $E[4](\bar{\mathbb{Q}}) \subseteq E(\mathbb{Q}(\zeta_8))$ . Compare with  $\mu_8(\bar{\mathbb{Q}})$  and Fact 3.1.

- (4) Show that  $\sigma$  restricts to a group automorphism of  $E[4](\overline{\mathbb{Q}})$ . Compute  $P_2 + (1, 0)$ , and write  $\sigma(P_2)$  and  $\sigma(Q_2)$  in the basis  $(P_2, Q_2)$ . Compare the determinant of the resulting matrix in  $\text{GL}_2(\mathbb{Z}/4\mathbb{Z})$  with the action of  $\sigma$  on  $\mu_4(\overline{\mathbb{Q}})$  found in (1), and explain the agreement using Fact 3.1.
- (5) Let  $(P_n)_n$  and  $(Q_n)_n$  be compatible sequences,  $[2]P_{n+1} = P_n$  and  $[2]Q_{n+1} = Q_n$ , starting with  $P_1 = (0, 0)$ ,  $Q_1 = (1, 0)$  and  $P_2, Q_2$  as above. Explain why they form a  $\mathbb{Z}_2$ -basis of  $T_2(E)$ , and compare with (1).

*Solution.*

(1) The map  $\zeta \mapsto \zeta^2$  sends the primitive  $2^{n+1}$ -th roots of unity onto the primitive  $2^n$ -th ones, so there are compatible sequences  $(\zeta_{2^n})_n$  of primitive roots with  $\zeta_{2^{n+1}}^2 = \zeta_{2^n}$ . Since  $\zeta_{2^n}$  generates the cyclic group  $\mu_{2^n}(\overline{\mathbb{Q}})$ , the maps  $\mathbb{Z}/2^n\mathbb{Z} \rightarrow \mu_{2^n}(\overline{\mathbb{Q}})$ ,  $a \mapsto \zeta_{2^n}^a$ , are isomorphisms compatible with reduction on the left and squaring on the right, and in the limit  $\mathbb{Z}_2 \xrightarrow{\sim} T_2(\mathbb{G}_m)$ . So  $(\zeta_{2^n})_n$  is a  $\mathbb{Z}_2$ -basis, for instance  $(-1, i, \zeta_8, \dots)$  with  $\zeta_8 = (1+i)/\sqrt{2} = e^{2\pi i/8}$ . Complex conjugation inverts every root of unity: on  $\mu_4(\overline{\mathbb{Q}})$  it is  $i \mapsto -i = i^3$ , and on  $T_2(\mathbb{G}_m)$  it is multiplication by  $-1 \in \mathbb{Z}_2^\times$ .

(2)  $E[2](\overline{\mathbb{Q}}) = \{O, (0, 0), (1, 0), (-1, 0)\}$ . A point  $P \in E(\overline{\mathbb{Q}})$  with  $[2]P \in E[2](\overline{\mathbb{Q}}) \setminus \{O\}$  is not in  $E[2](\overline{\mathbb{Q}})$ , so  $4(x_P^3 - x_P) = 4y_P^2 \neq 0$ . For such  $P$ , evaluating Problem 1.1(6) at  $P$  shows that  $x([2]P) = ([2]^*x)(P) = e$ , that is  $[2]P = (e, 0)$ , exactly when  $g_e(x_P) = 0$ : so  $x_P = \pm i$  for  $e = 0$ ,  $x_P = 1 \pm \sqrt{2}$  for  $e = 1$  and  $x_P = -1 \pm \sqrt{2}$  for  $e = -1$ . None of these is a root of  $x^3 - x$ , so each gives two points  $(x_P, \pm y_P) \in E(\overline{\mathbb{Q}})$  outside  $E[2](\overline{\mathbb{Q}})$ , twelve in all. With  $E[2](\overline{\mathbb{Q}})$  this makes 16 points of  $E(\overline{\mathbb{Q}})$ ; they are all of  $E[4](\overline{\mathbb{Q}})$ , since a point of  $E[4](\overline{\mathbb{Q}})$  not in  $E[2](\overline{\mathbb{Q}})$  doubles into  $E[2](\overline{\mathbb{Q}}) \setminus \{O\}$ .

(3) For  $P_2$ ,  $x_{P_2}^3 - x_{P_2} = i^3 - i = -2i = (1-i)^2 = y_{P_2}^2$ . For  $Q_2$ ,  $x_{Q_2}^2 = (1+\sqrt{2})^2 = 3+2\sqrt{2}$ , so  $x_{Q_2}^3 - x_{Q_2} = x_{Q_2}(x_{Q_2}^2 - 1) = 2x_{Q_2}^2 = 6+4\sqrt{2} = (2+\sqrt{2})^2 = y_{Q_2}^2$ . By (2),  $x([2]P_2) = 0$  and  $x([2]Q_2) = 1$ , and a point of  $E(\overline{\mathbb{Q}})$  with  $x$ -coordinate 0 or 1 has  $y$ -coordinate 0. If  $[a]P_2 + [b]Q_2 = O$ , doubling gives  $[a](0, 0) + [b](1, 0) = O$ , so  $a = 2a'$  and  $b = 2b'$  are even; then  $[a'](0, 0) + [b'](1, 0) = O$  makes  $a', b'$  even too. So  $(\mathbb{Z}/4\mathbb{Z})^2 \rightarrow E[4](\overline{\mathbb{Q}})$ ,  $(a, b) \mapsto [a]P_2 + [b]Q_2$ , is injective, hence bijective. Every point of  $E[4](\overline{\mathbb{Q}})$  thus lies in  $E(\mathbb{Q}(i, \sqrt{2}))$ , and  $\mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$ , since  $\zeta_8 = (1+i)/\sqrt{2}$ . This is exactly  $\mathbb{Q}(\mu_8(\overline{\mathbb{Q}}))$ . Writing  $\mathbb{Q}(E[4](\overline{\mathbb{Q}}))$  for the field generated by the coordinates of the points of  $E[4](\overline{\mathbb{Q}})$ , Fact 3.1 forces  $\mathbb{Q}(E[4](\overline{\mathbb{Q}})) \supseteq \mathbb{Q}(\mu_4(\overline{\mathbb{Q}}))$ : the values of  $e_4$  lie in  $\mathbb{Q}(E[4](\overline{\mathbb{Q}}))$  and include a primitive fourth root of unity.

(4) The group law on  $E(\overline{\mathbb{Q}})$  is given by rational functions with coefficients in  $\mathbb{Q}$ , so  $\sigma(P+Q) = \sigma(P) + \sigma(Q)$ ; in particular  $[4]\sigma(P) = \sigma([4]P)$ , and  $\sigma$  maps  $E[4](\overline{\mathbb{Q}})$  to itself, with inverse  $\sigma$ . Let  $R := P_2 + (1, 0)$ . The line through  $(i, 1-i)$  and  $(1, 0)$  has slope  $\lambda = -1$ , so  $x_R = \lambda^2 - i - 1 = -i$  and  $y_R = -((1-i) + \lambda(x_R - i)) = -1 - i$ :  $R = (-i, -1-i)$ . Hence  $\sigma(P_2) = (-i, 1+i) = -R = -P_2 - [2]Q_2$ . Since  $P_2$  and  $Q_2$  have order 4, this is  $[3]P_2 + [2]Q_2$ . And  $\sigma(Q_2) = Q_2$ , since its coordinates are real. In the basis  $(P_2, Q_2)$ ,  $\sigma|_{E[4](\overline{\mathbb{Q}})}$  has matrix

$$\begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}, \quad \det = 3 \equiv -1 \pmod{4},$$

and by (1),  $\sigma$  acts on  $\mu_4(\overline{\mathbb{Q}})$  by the same exponent 3. This is no accident: by Fact 3.1,  $e_4(P_2, Q_2)$  is a primitive fourth root of unity and  $\sigma(e_4(P_2, Q_2)) = e_4(P_2, Q_2)^{\det} = e_4(P_2, Q_2)^3$ .

(5) The map  $(\mathbb{Z}/2^n\mathbb{Z})^2 \rightarrow E[2^n](\overline{\mathbb{Q}})$ ,  $(a, b) \mapsto [a]P_n + [b]Q_n$ , is an isomorphism for every  $n$  (the argument of (3) by induction), and these isomorphisms are compatible with reduction on the

left and [2] on the right. Passing to the limit,  $\mathbb{Z}_2^2 \xrightarrow{\sim} T_2(E)$ . This is (1) in rank two:  $\mu_{2^n}(\overline{\mathbb{Q}})$  is free of rank one over  $\mathbb{Z}/2^n\mathbb{Z}$ , while  $E[2^n](\overline{\mathbb{Q}})$  is free of rank two.

#### 4. MAGMA EXAMPLES

**Example 4.1** (The isogenies of Problem 2.1). We check in the function field of  $E_1$  that  $\varphi$  and  $\psi$  land where they should, that  $\iota \circ \psi \circ \varphi = [2]$ , and compare with Magma's own 2-isogeny.

---

```

E := EllipticCurve([-1,0]);           // y^2 = x^3 - x
Kx<x> := FunctionField(Rationals());
R<t> := PolynomialRing(Kx);
F<y> := ext<Kx | t^2 - (x^3 - x)>;      // x = x_1, y = y_1
x2 := F!((x^2 - 1)/x); y2 := y*(x^2 + 1)/x^2; // phi^*x_2, phi^*y_2
y2^2 eq x2^3 + 4*x2;                  // true
x3 := y2^2/x2^2; y3 := y2*(x2^2 - 4)/x2^2; // (psi o phi)^*x_3, ^*y_3
y3^2 eq x3^3 - 16*x3;                 // true
lam := (3*x^2 - 1)/(2*y); xd := lam^2 - 2*x; yd := -(y + lam*(xd - x));
x3/4 eq xd and y3/8 eq yd;            // true: [2]
phi := TwoIsogeny(E![0,0]);
aInvariants(Codomain(phi));           // [ 0, 0, 0, 4, 0 ]
phi(E![1,0]);                         // (0 : 0 : 1)
DualIsogeny(phi);
// ... taking (x : y : 1) to ((1/4*x^2 + 1) / x : (1/8*x^2*y - 1/2*y) / x^2 : 1)

```

---

Magma's dual is  $\iota \circ \psi$  written out in the coordinates  $x_2, y_2$  of  $E_2$ :  $((x_2^2 + 4)/(4x_2), y_2(x_2^2 - 4)/(8x_2^2))$ .

**Example 4.2** (Division polynomials and  $E[4](\overline{\mathbb{Q}})$ ). For  $m \geq 2$ ,  $\text{DivisionPolynomial}(E, m)$  vanishes exactly at the  $x$ -coordinates of  $E[m](\overline{\mathbb{Q}}) \setminus \{O\}$ . Its degree is  $(m^2 - 1)/2$  for odd  $m$  and  $(m^2 + 2)/2$  for even  $m$ , and its factors for  $m = 4$  are  $x, x \pm 1$  and the  $g_e$  of Problem 1.1(6).

---

```

[ Degree(DivisionPolynomial(E,m)) : m in [2..8] ];
// [ 3, 4, 9, 12, 19, 24, 33 ]
Factorization(DivisionPolynomial(E,4));
// x - 1, x, x + 1, x^2 - 2*x - 1, x^2 + 1, x^2 + 2*x - 1
Q8<z> := CyclotomicField(8); i := z^2; s := z + z^7; // s^2 = 2
E8 := BaseChange(E, Q8);
P2 := E8![i, 1-i]; Q2 := E8![1+s, 2+s];
2*P2, 2*Q2; // (0 : 0 : 1) (1 : 0 : 1)
#DivisionPoints(E8!0, 4); // 16
E8![-i, 1+i] eq 3*P2 + 2*Q2; // true: complex conjugation

```

---

**Example 4.3** (Characteristic  $p$ ). Reducing modulo  $p$ , the  $p$ -division polynomial collapses, because  $\#E[p](\mathbb{F}_p) \in \{1, p\}$ .

---

```

for p in [3,5,7,13] do
  Ep := ChangeRing(E, GF(p));
  psi := DivisionPolynomial(Ep, p);
  print p, IsSupersingular(Ep), Degree(psi), Factorization(psi);
end for;
// 3 true 0 []
// 5 false 10 [ <x^2 + 2, 5> ]

```

---

```
// 7 true 0 []
// 13 false 78 [ <x^6 + 11*x^4 + 5*x^2 + 11, 13> ]
```

---

For  $p = 3, 7$  the curve is supersingular and  $E[p](\overline{\mathbb{F}}_p) = 0$ ; in general,  $y^2 = x^3 - x$  is supersingular exactly when  $p \equiv 3 \pmod{4}$ . For  $p = 5, 13$  the division polynomial is the  $p$ -th power of a polynomial of degree  $(p-1)/2$ , whose roots are the  $x$ -coordinates of the  $p-1$  non-zero points of  $E[p](\overline{\mathbb{F}}_p) \cong \mathbb{Z}/p\mathbb{Z}$ ; the exponent  $p$  is the inseparable degree of  $[p]$ . The same computation for  $\mathbb{G}_m$  is  $t^p - 1 = (t - 1)^p$ .

**Looking ahead.** The Weil pairing of Fact 3.1 also gives  $\det \varphi_\ell = \deg \varphi$  for the map  $\varphi_\ell$  induced by an isogeny on Tate modules ([Sil09, Proposition III.8.6]); formal groups make  $c_\varphi$  an honest derivative; and Frobenius factors  $[p]$  in characteristic  $p$ .

## REFERENCES

- [Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑